THE CENTER OF A LOCALLY ALGEBRAIC GROUP SCHEME IS A CLOSED SUBGROUP SCHEME

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file groupoids.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: this chapter's pool has 47 of 128 labels already delivered by the S7 line (S7-gp-* registrations, statement_anchor-prose scan; withdrawn cards excluded); none of the 11 accepted labels of this batch is among them (gp-occupied.json, 2026-10-14).

Source reference. Original reference 06LH: sources/stacks-chow/tags.txt, source lines 8466–8466. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 022W: sources/stacks-chow/tags.txt, source lines 2721–2721. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00IR: sources/stacks-chow/tags.txt, source lines 689–689. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original proposition 0B7R: sources/stacks-chow/tags.txt, source lines 14114–14114. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0B7P: sources/stacks-chow/tags.txt, source lines 14112–14112. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03I0: sources/stacks-chow/tags.txt, source lines 4586–4586. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02L6: sources/stacks-chow/tags.txt, source lines 3397–3397. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let k be a field. Let G be a locally algebraic group scheme over k. Then the center of G is a closed subgroup scheme of G.*

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Proof. Let $\mathrm{Aut}(G)$ denote the contravariant functor on the category of schemes over k which associates to S/k the set of automorphisms of the base change G_S as a group scheme over S. There is a natural transformation

$$G \longrightarrow \mathrm{Aut}(G), \quad g \longmapsto \mathrm{inn}_g$$

sending an S-valued point g of G to the inner automorphism of G determined by g. The center C of G is by definition the kernel of this transformation, i.e., the functor which to S associates those $g \in G(S)$ whose associated inner automorphism is trivial. The statement of the lemma is that this functor is representable by a closed subgroup scheme of G.

Choose an integer $n \geq 1$. Let $G_n \subset G$ be the nth infinitesimal neighbourhood of the identity element e of G. For every scheme S/k the base change $G_{n,S}$ is the nth infinitesimal neighbourhood of $e_S : S \to G_S$. Thus we see that there is a natural transformation $\mathrm{Aut}(G) \to \mathrm{Aut}(G_n)$ where the right hand side is the functor of automorphisms of G_n as a scheme (G_n isn't in general a group scheme). Observe that G_n is the spectrum of an artinian local ring A_n with residue field k which has finite dimension as a k-vector space (Varieties, Lemma 06LH). Since every automorphism of G_n induces in particular an invertible linear map $A_n \to A_n$, we obtain transformations of functors

$$G \to \mathrm{Aut}(G) \to \mathrm{Aut}(G_n) \to \mathrm{GL}(A_n)$$

The final group valued functor is representable, see Example 022W, and the last arrow is visibly injective. Thus for every n we obtain a closed subgroup scheme

$$H_n = \mathrm{Ker}(G \to \mathrm{Aut}(G_n)) = \mathrm{Ker}(G \to \mathrm{GL}(A_n)).$$

As a first approximation we set $H = \bigcap_{n \geq 1} H_n$ (scheme theoretic intersection). This is a closed subgroup scheme which contains the center C.

Let h be an S-valued point of H with S locally Noetherian. Then the automorphism inn_h induces the identity on all the closed subschemes $G_{n,S}$. Consider the kernel $K = \mathrm{Ker}(\mathrm{inn}_h : G_S \to G_S)$. This is a closed subgroup scheme of G_S over S containing the closed subschemes $G_{n,S}$ for $n \geq 1$. This implies that K contains an open neighbourhood of $e(S) \subset G_S$, see Algebra, Remark 00IR. Let $G^0 \subset G$ be as in Proposition 0B7R. Since G^0 is geometrically irreducible, we conclude that K contains G^0_S (for any nonempty open $U \subset G^0_{k'}$ and any field extension k'/k we have $U \cdot U^{-1} = G^0_{k'}$, see proof of Lemma 0B7P). Applying this with $S = H$ we find that G^0 and H are subgroup schemes of G whose points commute: for any scheme S and any S-valued points $g \in G^0(S)$, $h \in H(S)$ we have $gh = hg$ in $G(S)$.

Assume that k is algebraically closed. Then we can pick a k-valued point g_i in each irreducible component G_i of G. Observe that in this case the connected components of G are the irreducible components of G are the translates of G^0 by our g_i. We claim that

$$C = H \cap \bigcap\nolimits_i \mathrm{Ker}(\mathrm{inn}_{g_i} : G \to G) \quad \text{(scheme theoretic intersection)}$$

Namely, C is contained in the right hand side. On the other hand, every S-valued point h of the right hand side commutes with G^0 and with g_i hence with everything in $G = \bigcup G^0 g_i$.

The case of a general base field k follows from the result for the algebraic closure $\overline{k}$ by descent. Namely, let $A \subset G_{\overline{k}}$ the closed subgroup scheme representing the center of $G_{\overline{k}}$. Then we have

$$A \times_{\mathrm{Spec}(k)} \mathrm{Spec}(\overline{k}) = \mathrm{Spec}(\overline{k}) \times_{\mathrm{Spec}(k)} A$$

as closed subschemes of $G_{\overline{k}\otimes_k\overline{k}}$ by the functorial nature of the center. Hence we see that A descends to a closed subgroup scheme $Z \subset G$ by Descent, Lemma 03I0 (and Descent, Lemma 02L6). Then Z represents C (small argument omitted) and the proof is complete. □